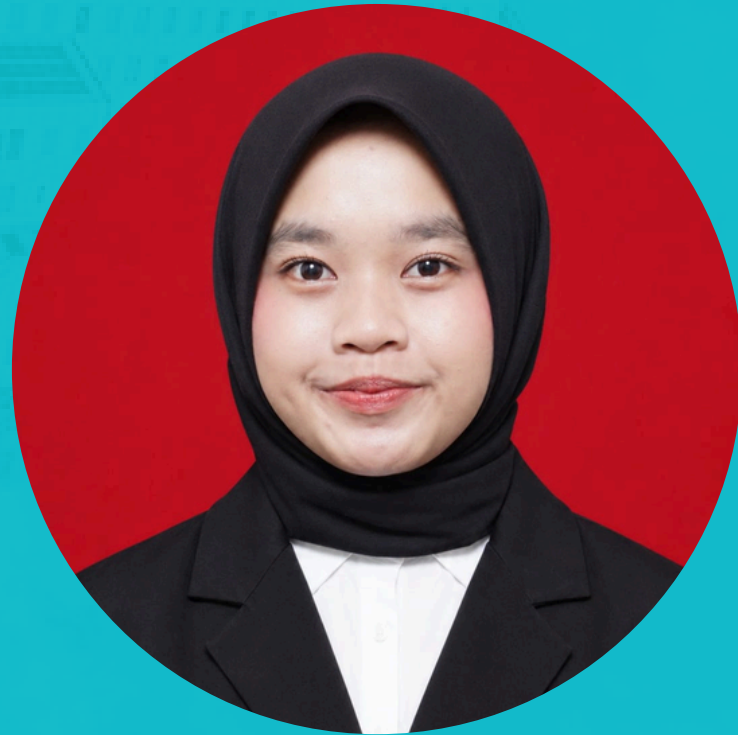


Sales Performance Dashboard

Business Intelligence Analyst
Bank Muamalat × Rakamin Academy

Presented by Anisah Nabilah Ghasani



Anisah Nabilah Ghasani

About Me

Fresh graduate in Statistics from Universitas Airlangga with an interest in Data Analytics and Business Intelligence. Experienced in data processing, analysis, and reporting through internship and research experience. Familiar with Microsoft Excel, SQL, Python, Tableau, and R, and interested in using data to generate insights and support business decision-making.

My Experience

Research Intern Universitas Airlangga | Jul – Dec 2025

- Collected, prepared, and analyzed panel data from 48 Asian countries covering 2020–2023 to examine factors associated with CO₂ emissions.
- Developed and evaluated 3,276 model alternatives using Generalized Cross Validation (GCV) to identify the optimal model.
- Interpreted model results and thematic maps to identify differences in predictor effects across countries.

Planning Division Internship PT PLN (Persero) UID Jawa Timur | Jun – Jul 2024

- Validated operational performance data across 16 PLN service units and supported internal performance evaluation.
- Managed KPI datasets and analyzed 138 monthly electricity price observations to identify trends and generate insights for planning.
- Supported reporting, risk monitoring, and strategic planning activities.

Company Overview

PT Bank Muamalat Indonesia Tbk is the first Islamic bank in Indonesia, established on November 1, 1991, based on an initiative from the Indonesian Ulema Council (MUI), the Indonesian Muslim Intellectuals Association (ICMI), and Muslim entrepreneurs, with support from the Indonesian Government. The Bank began its operations on May 1, 1992, and has since played a pioneering role in the development of Islamic banking and financial services in Indonesia.

Through this Business Intelligence Analyst Project-Based Internship with Rakamin Academy, the project focuses on applying data processing, analysis, and visualization to generate insights that can support data-driven business decisions.

Data Analytics

Business Intelligence

Data Visualization

Business Insights

Case Study

PT Sejahtera Bersama

PT Sejahtera Bersama is a company with transaction data covering customer information, orders, products, and product categories. In this project, I take the role of a Business Intelligence Analyst to transform transaction data into meaningful insights that can support sales performance evaluation and business decision-making.

Project Objectives

Data Integration → Combining customer, order, product, and category data into a structured dataset.

Data Analysis → Analyzing sales performance and transaction quantity across categories and cities.

Data Visualization → Developing an interactive dashboard to present key sales metrics and trends.

Business Insights → Identifying sales opportunities and providing data-driven recommendations.

Data Source

<https://drive.google.com/file/d/1RwsBQ1FriNfz6qiq0V5nD7gF7jO81To3/view>

Question 1

Primary Key

Identify the Primary Key of Each Table

Table	Primary Key
Customers	CustomerID
Products	ProdNumber
Orders	OrderID
Product Category	CategoryID

Customer ID → Identifies each customer and distinguishes one customer record from another.

ProdNumber → Serves as the unique product code for distinguishing each product in the dataset.

OrderID → Distinguishes each order or transaction, allowing every transaction to be identified individually.

CategoryID → Distinguishes one product category from another and identifies each category in the dataset.

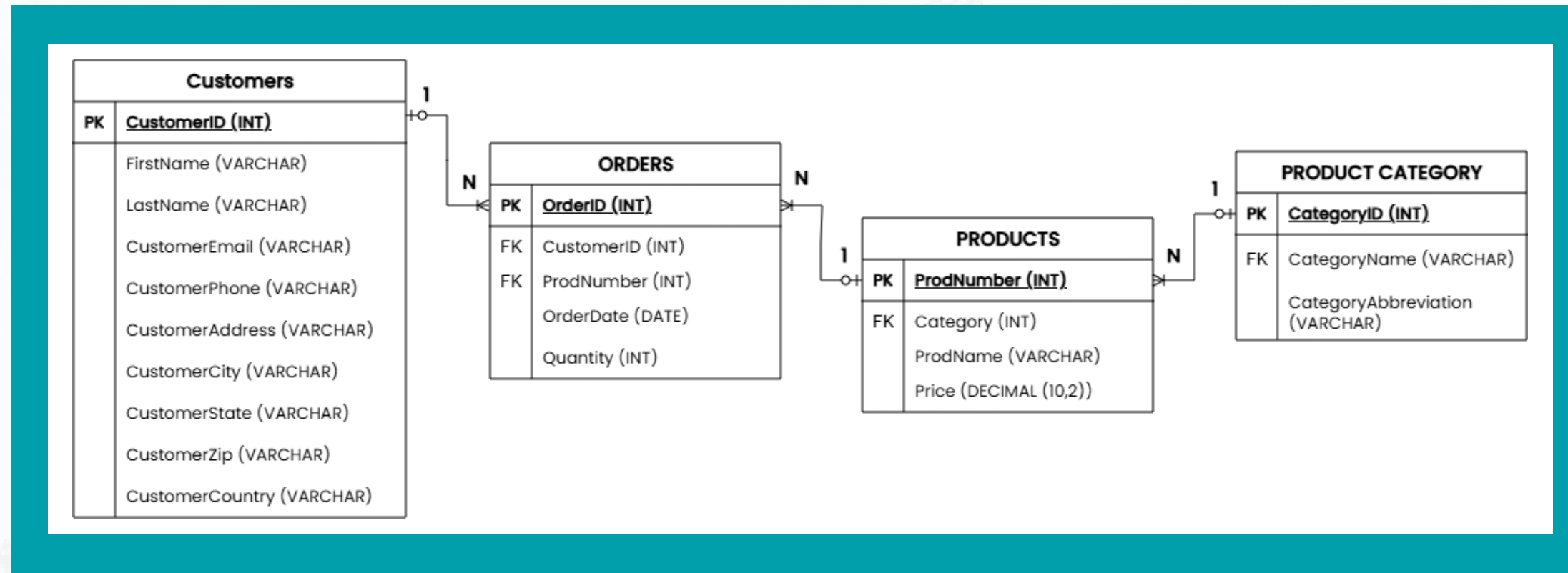
Why Primary Keys ?

A primary key is a column used as a unique identifier to distinguish each record within a table.

Question 2

Database Relationship

Identify the Relationship between Each Table



Customers → Orders

CustomerID → CustomerID | One-to-Many (1:N)

One customer can have multiple orders.

Products → Orders

ProdNumber → ProdNumber | One-to-Many (1:N)

One product can appear in multiple orders.

Product Category → Products

CategoryID → Category | One-to-Many (1:N)

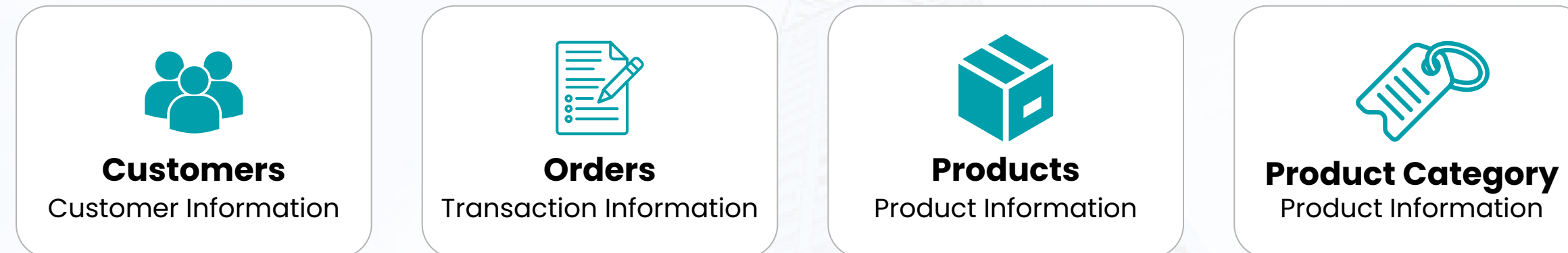
One category can contain multiple products.

Question 3

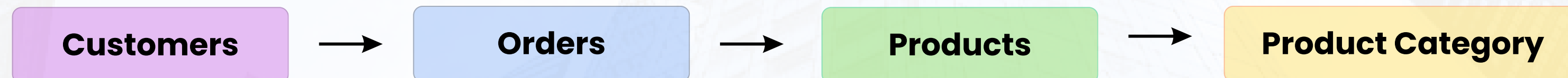
Creating on Integrated Transaction Table

Combine customer, transaction, product, and product category information into a single transaction-level dataset

Data Sources



Data Integration



Join Relationships

Customers.Customerid → Orders.CustomerID
Orders.ProdNumber → Products.ProdNumber
Products.Category → ProductCategory.CategoryID

Key Calculation



TotalSales = OrderQty × ProductPrice

Results

Combine customer, transaction, product, and product category information into a single transaction-level dataset

Question 3

Creating MasterTable using Google BigQuery

The following SQL query is used to create the MasterTable by joining the four datasets based on their relationships.

SQL Query (Google BiQuery)

Standard SQL

Query Explanation

```
1 SELECT
2   c.CustomerEmail AS CustomerEmail,
3   c.CustomerCity AS CustomerCity,
4   o.Date AS OrderDate,
5   o.Quantity AS OrderQty,
6   p.ProdName AS ProductName,
7   p.Price AS ProductPrice,
8   pc.CategoryName AS ProductCategoryName,
9   o.Quantity * p.Price AS TotalSales
10 FROM
11   `project-dc175103-acb0-45f5-bdc.FinalTask_BankMuamalat.Customers` AS c
12 JOIN
13   `project-dc175103-acb0-45f5-bdc.FinalTask_BankMuamalat.Orders` AS o
14   ON c.CustomerID = o.CustomerID
15 JOIN
16   `project-dc175103-acb0-45f5-bdc.FinalTask_BankMuamalat.Products` AS p
17   ON o.ProdNumber = p.ProdNumber
18 JOIN
19   `project-dc175103-acb0-45f5-bdc.FinalTask_BankMuamalat.ProductCategory` AS pc
20   ON p.Category = pc.CategoryID
21 ORDER BY
22   o.Date ASC;
```

JOIN

Combines the four datasets based on their relationships.

CALCULATION

TotalSales = OrderQty × ProductPrice

Calculates the total sales for each transaction

SORTING

Order By = OrderDate ASC

Transactions are ordered by OrderDate in ascending order, from the earliest to the latest.

Question 4

MasterTable Result in Google BigQuery

The query successfully created a MasterTable that combines customer, transaction, product and product category information.

☆	MasterTable	Query	Chat	Open in ▾	Share ▾	Copy	Snapshot	Delete	Export / sync ▾	Re
Schema	Details	Preview	Insights	Lineage	Data Profile	Data Quality				
Row	CustomerEmail	CustomerCity	OrderDate	OrderQty	ProductName	ProductPrice	ProductCategoryName	TotalSales		
1	gmacenzy8g@constantcontact.com#mailto:gmacenzy8g@constantcontact.com#	Springfield	2020-02-01	1	AI for Educators	49.95	Training Videos	49.95		
2	easkella0@eventbrite.com#mai...	New Orleans	2020-04-03	1	AI for Educators	49.95	Training Videos	49.95		
3	rcouplandgz@google.com.br#...	San Antonio	2020-05-21	1	AI for Educators	49.95	Training Videos	49.95		
4	rslocombbh@wunderground.com#mailto:rslocombbh@wunderground.com#	Appleton	2020-05-23	1	AI for Educators	49.95	Training Videos	49.95		
5	cperrittov@homestead.com#m...	Atlanta	2020-06-02	1	AI for Educators	49.95	Training Videos	49.95		
6	rchristoffelw@so-net.ne.jp#mailto:rchristoffelw@so-net.ne.jp#	Baltimore	2020-07-27	1	AI for Educators	49.95	Training Videos	49.95		
7	rcharville3k@ovh.net#mailto:rc...	Anniston	2021-08-09	1	AI for Educators	49.95	Training Videos	49.95		
8	ilethyib@ustream.tv#mailto:ile...	Brooklyn	2021-12-17	1	AI for Educators	49.95	Training Videos	49.95		
9	xhulle6v@shinystat.com#mailt...	Charlotte	2020-01-05	2	AI for Educators	49.95	Training Videos	99.9		
10	ttoulchi5@ehow.com#mailto:tt...	Phoenix	2020-02-10	2	AI for Educators	49.95	Training Videos	99.9		
11	rbevirnj@blogspot.com#mailto:...	Houston	2020-04-05	2	AI for Educators	49.95	Training Videos	99.9		
12	wlyness2x@twitpic.com#mailto...	San Diego	2020-07-05	2	AI for Educators	49.95	Training Videos	99.9		
13	ehamprecht18@dot.gov#mailto...	Nashville	2020-10-02	2	AI for Educators	49.95	Training Videos	99.9		

MasterTable Information

Table Name : **MasterTable**
Dataset : FinalTask_Bank Muamalat
Platform : Google BigQuery

MasterTable Columns

- CustomerEmail
- CustomerCity
- OrderDate
- OrderQty
- ProductName
- ProductPrice
- Product CategoryName
- TotalSales

Question 4

Data Visualization



Objective

Create an interactive dashboard using Looker Studio to visualize sales performance, customer behavior, product performance, and city-level insights based on the **MasterTable**.



Data Source

MasterTable (Google BigQuery)

The dashboard uses the MasterTable created in Question 3, which contains customer, transaction, product, and product category information.



Tool

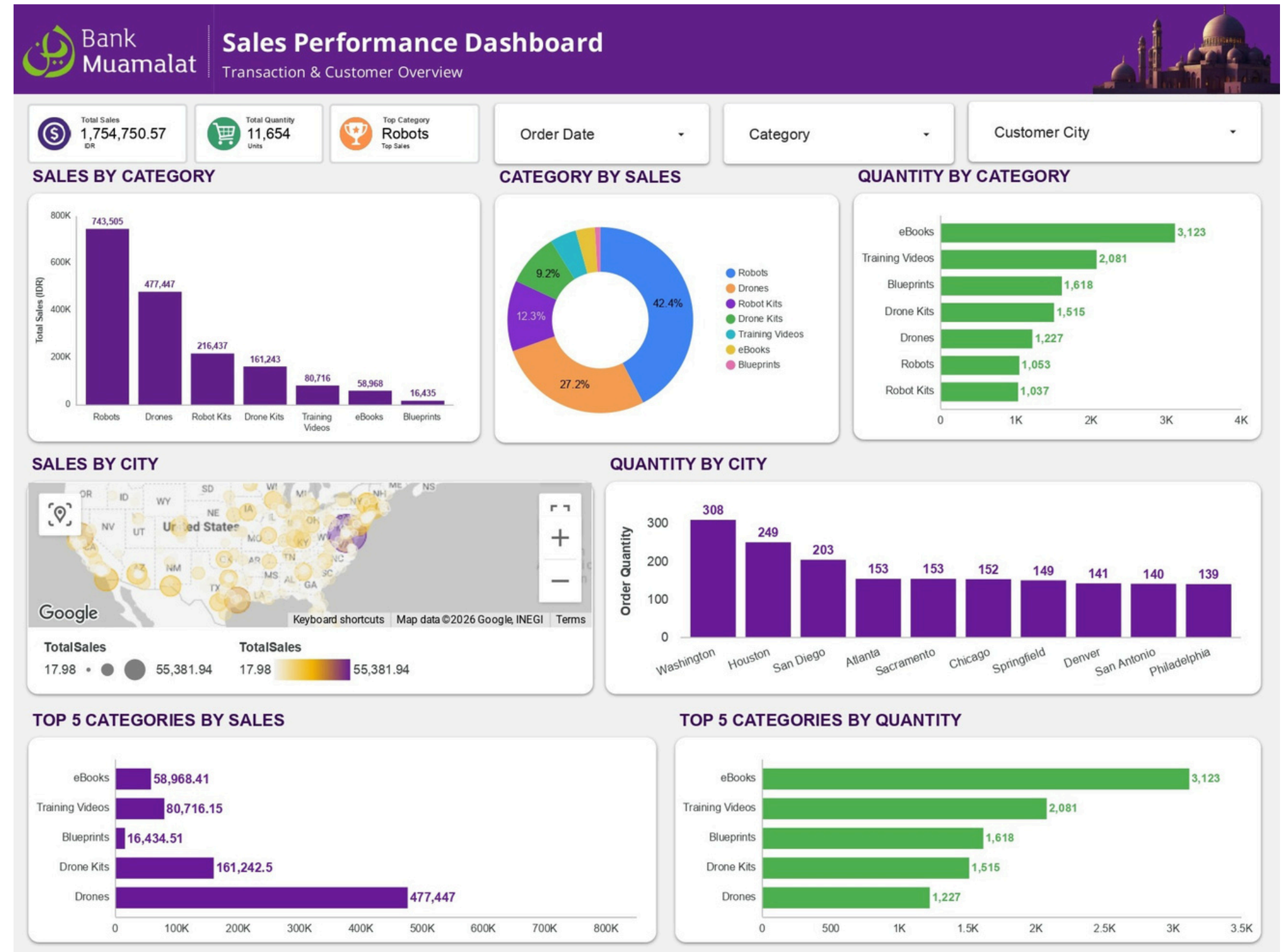
Looker Studio

Used to create an interactive dashboard from the **MasterTable**.



Visualization Included

- Total Overall Sales
- Total Sales by Product Category
- Total Quality by Product Category
- Total Sales by City
- Total Quality by City
- Top 5 Product Categories by Sales
- Top 5 Product Categories by Quantity



[Dashboard Looker Studio](#)



Question 5

Sales Strategy Recommendation

Turning Transaction Insights into Business Actions



Maintain High-Performing Categories

Focus on Robots and Drones by maintaining product availability, quality, and relevant promotions.



Robots
IDR 743,505



Drones
IDR 477,447



Bundling & Cross-Selling

Bundle high-sales products with complementary products, such as Robots+ Robot Kits/Training Videos and Drones + Drone Kits/Training Videos to increase transaction value.



Leverage High-Quantity Products

Use eBooks and Training Videos as complementary products for customers purchasing higher-value products (Robots/Drones) to drive additional sales.



Improve Low-Sales Categories

Evaluate and promote categories such as Blueprints and eBooks through targeted promotions, bundling, or product recommendations to increase sales performance.



Retain High-Activity Cities

Target Washington, Houston, and San Diego with loyalty programs, special promotions, and relevant offers to encourage repeat purchases, as these cities recorded the highest transaction quantities.



Washington
308 units



Houston
249 units



San Diego
203 units

Key Takeaway

Maintain what performs well, increase transaction value through cross-selling, and develop products and markets with growth potential.

Video Presentation

[Link Video Presentasi Final Task Rakamin Academy x Bank Muamalat](#)

Thank You



Bank
Muamalat

